

Ziwen (Aaron) Li

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RESEARCH INTERESTS

My research focuses on **adaptive agent orchestration** and **privacy-preserving AI**. I study how agents learn to acquire information, coordinate reasoning, and delegate tasks across collaborators with different capabilities, under privacy and resource constraints.

EDUCATION & RESEARCH EXPERIENCE

- **Northeastern University** Boston, MA
Ph.D. in Computer Science
Advisor: Prof. Tianshi Li
2025 – present
- **University of California, Los Angeles** Los Angeles, CA
M.S. in Electrical and Computer Engineering (Thesis Track)
Graduate Student Researcher; Advisors: Prof. Xiang ‘Anthony’ Chen, Prof. Eunice Jun
2023 – 2025
- **Westlake University** Hangzhou, China
Visiting Student Researcher; Advisor: Prof. Yue Zhang
Spring 2023
- **University of Electronic Science and Technology of China (UESTC) & University of Glasgow** Chengdu, China
Joint B.E. (Honours)
Electronic Information Engineering (UESTC); Electronics and Electrical Engineering (Glasgow)
2019 – 2023

SELECTED PUBLICATIONS

- **LLM Anonymization Against Agentic Re-Identification** [[Paper](#)] [[Code](#)]
Ziwen Li, Jianing Wen, and Tianshi Li. *COLM HAIPS Workshop 2026*.
 - Led **AURA** to protect interview transcripts from agentic re-identification, introducing adaptive privacy scopes and a mask–reconstruct pipeline that preserves downstream context.
 - Evaluated 53 high-risk interviews against three web-search attackers, with 4,758 validated contextual utility units.
 - Hybrid Qwen3.5-35B-A3B raised evidence-supported utility recovery from **66.3% to 76.1%** and reduced GPT-5.4-mini re-identifications from **12 to 7** versus the prior LLM anonymizer.
- **Planning over Private State: Local–Remote Delegation for Tool-Calling Agents**
Ziwen Li and Tianshi Li. 2026. *In Submission*.
 - Leading **DAM-Guard**, separating remote planning from local private-state computation for tool-calling agents.
 - Implemented three local–remote protocols: natural-language queries, structured queries, and proactive evidence.
 - Introduced **computation placement as a data-minimization mechanism**: remote agents request task-relevant evidence from a local agent through a controlled delegation interface.
- **RAGPPI: Retrieval-Augmented Generation Benchmark for Protein–Protein Interactions in Drug Discovery** [[Paper](#)] [[Dataset](#)]
Youngseung Jeon, Ziwen Li, Thomas Li, JiaSyuan Chang, Morteza Ziyadi, and Xiang ‘Anthony’ Chen. *EACL 2026*.
 - Implemented automated factuality evaluation for scientific RAG, combining atomic-fact verification, embedding similarity, and majority voting across three LLM judges.
 - Curated **3,720 silver-standard QA pairs** using agreement between two evidence-grounded judges.
 - Reduced evaluation error from **8.57% to 4.29%** versus the best individual judge on 70 expert-labeled examples (**50% fewer errors**).
- **GraPPI: A Retrieve-Divide-Solve GraphRAG Framework for Large-scale Protein-protein Interaction Exploration** [[Paper](#)]
Ziwen Li, Xiang ‘Anthony’ Chen, and Youngseung Jeon. *Findings of NAACL 2025*.
 - Led **GraPPI**, introducing a retrieve–divide–solve GraphRAG pipeline for task-relevant context acquisition.
 - Combined FAISS retrieval over **2.96M protein interactions** with parallel edge-level reasoning, condensing evidence into context for pathway explanation and reranking.
 - Raised GPT-4o ROUGE-L from **30.06 to 35.67 (+5.61 points)** over RAG without CoT.

SERVICE

- **Invited Talk**: Tufts University for 3rd Annual Boston Security Usability Research Day (ABSURD 2026).
- **Reviewer**: CHI, COLM.